

Immersive Virtual Reality for Vocational Training: A Case Study in Bakery Sales Training

Forouzan Farzinnejad*

Navid Khezrian†

Seyedmasih Tabaei‡

Jens Grubert§

Coburg University of Applied Sciences and Arts

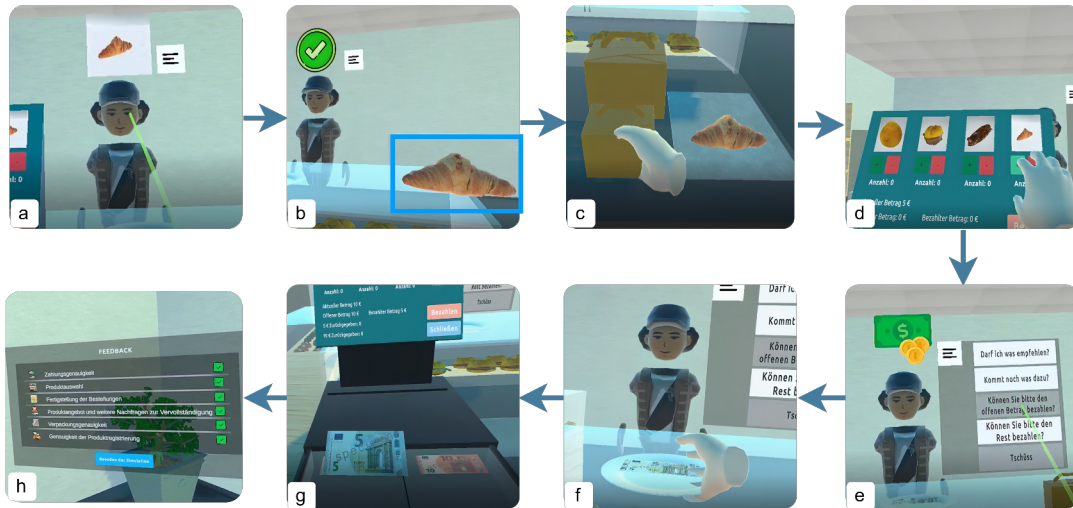


Figure 1: Overview of the VR Bakery Sales Training Sequence. a) Customer interaction b) Product suggestion c) Product packaging d) Recording of product prices e-g) Initiation of payment process h) Functional feedback.

ABSTRACT

This study presents a Virtual Reality vocational training system for bakery sales, offering a potentially cost-effective and immersive solution to traditional training methods. The VR platform is designed to simulate realistic customer interactions, product handling, and transactions, which are believed to support trainees in developing skills, building confidence, and preparing for real-world tasks. A study with ten participants yielded a System Usability Scale score of 72.5, indicating positive usability feedback. This work explores VR's potential for vocational training through immersive simulations that support skill development, confidence-building, and real-world readiness.

Index Terms: virtual reality, real-world task simulation, vocational training system, immersive learning

1 INTRODUCTION

The rapid progress of immersive technologies has opened up exciting possibilities for improving vocational education and training (VET). While traditional training methods remain effective, they are often constrained by being resource-intensive, expensive, and time-consuming. In comparison, augmented reality (AR) and virtual reality (VR) provide immersive, interactive, and potentially more economical alternatives [2, 5]. Moreover, the educa-

tional and entertainment potential of computer technology, particularly through virtual environments, can greatly enhance the teaching of practical skills [3]. VET, designed to equip learners with occupation-specific knowledge and competencies, is particularly well-suited to action-oriented and experiential learning models [2]. Empirical studies indicate that VR can significantly improve learning outcomes, making it an ideal tool for skill-based training [4, 6]. The action-oriented nature of VR aligns closely with the needs of vocational education, where practical, hands-on skills are prioritized. Furthermore, VR is widely accepted among learners, who report high usability and minimal discomfort, even as first-time users [4]. This acceptance indicates VR's potential as an effective medium for professional training in VET, enhancing both skill acquisition and learner engagement. Our research evolves around the question on how to improve vocational training in bakery sales. To this end, We propose a VR-based system by simulating realistic customer interactions, product handling, and transaction tasks. We evaluated the system through a descriptive user study to assess its usability, user experience, and perceived effectiveness in supporting skill acquisition and retention in a virtual training environment.

2 SYSTEM DESIGN AND IMPLEMENTATION

The training environment was designed to facilitate foundational skill acquisition for novice trainees in bakery sales through repetitive, structured practice within a virtual simulation. The program emphasizes core competencies, including customer service and transaction management, essential for proficiency in bakery sales roles. This simulated setting allows trainees to develop confidence and task fluency before engaging in real-world customer interactions.

In the VR system design, trainees experience typical customer transactions through interactive sequences (Fig. 1 for an overview).

*e-mail: forouzan.farzinnejad@hs-coburg.de

†e-mail: navid.khezrian@hs-coburg.de

‡e-mail: Seyedmasih.Tabaei@stud.hs-coburg.de

§e-mail: jens.grubert@hs-coburg.de

It involves identifying customer needs, selecting appropriate products, ensuring accurate packaging, processing payments, handling cash transactions, and a final functional feedback phase. The environment includes a virtual bakery, an infinite bread basket, and packaging options. Simulation customers make specific requests and initiate actions such as selecting, offering, and packaging. Afterward, trainees record transactions at the cash register and handle payments. The system uses a fixed price of €5 per item to streamline transaction management, directing trainee focus toward procedural accuracy. Throughout the training process, trainees will gain skills in customer service and transaction accuracy, while the functional feedback phase shows how they perform, highlighting correct actions and identifying errors.

The VR training environment was developed using the Unity game engine (version 2022.3.5f1) and the XR Interaction Toolkit, a framework designed for implementing VR interactions. The application was deployed on Meta Quest 3 head-mounted displays, with hand tracking and interaction facilitated by the Meta Quest 3 controllers.

Overall, this VR training system, in its initial development stage, provides an immersive environment for building essential bakery sales skills. It helps trainees gain practical experience, confidence, and accuracy by simulating customer interactions and transactions.

3 EVALUATION

The VR training system was evaluated with ten participants, each with prior experience in extended reality (XR) technology. The participant group consisted of nine males and one female, with a mean age of 30.9 years ($SD = 8.82$). Each participant volunteered as an investigator in the study, providing insights into both the usability and user experience of the VR system.

The evaluation process began with a brief overview of the VR system's purpose and an explanation of the tasks involved. Following this introduction, participants perform a complete set of VR training steps, which consisted of eight distinct steps, each performed only once and involving interaction with a single customer. Upon completing the session, participants provided qualitative feedback on their experience and rated the VR system's usability using the System Usability Scale (SUS) [1]. An IRB approval was not needed for this study.

Each session, including the briefing, VR task performance, and feedback collection, lasted approximately 20 minutes per participant. The qualitative feedback highlighted several positive aspects of the VR system, specifically regarding its user-friendliness, structured learning process, and effectiveness in skill development. The qualitative feedback indicated that: 1) The system's responsiveness was identified as a positive attribute, enhancing users' sense of immersion within the virtual environment. One participant noted, "The system is very responsive, so I felt immersed in the world in the model." 2) Participants generally found the application to be user-friendly and intuitive, with one noting, "It was pretty easy and straightforward." 3) Some participants acknowledged the application's effectiveness for training but suggested enhancements, particularly in simulating real-world interactions. One participant remarked, "It would be great if voice processing and interaction with voice were implemented." 4) The application was viewed as especially beneficial for younger users; however, concerns were raised regarding its accessibility for older adults, with one participant stating, "For an old person, it's not so easy but for younger participants, it's a good way to learn." 5) Participants suggested reference aids in the shop to clarify the function of each control for certain tasks. One participant suggested, "For some tasks, it would be helpful to have a clearer understanding of each control's function. A reference card displayed in the shop could provide this guidance."

Figure 2 illustrates the SUS score distribution, with a median of 73.75 and a mean of 72.5, indicating a central tendency around this

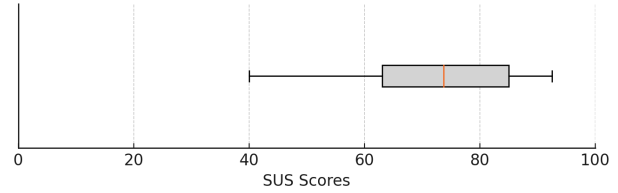


Figure 2: Boxplot of the system usability scale scores.

range. Most of the scores are concentrated between 65 and 85, as indicated by the interquartile range (IQR). While a few lower scores (e.g., 40.0) highlight areas for improvement, the standard deviation of 16.62 reflects moderate variability among participant responses, likely due to differences in participant familiarity or preferences.

4 CONCLUSION AND FUTURE WORK

This study presents a VR-based training system developed to enhance vocational training in bakery sales. By simulating customer interactions, product handling, and transaction processes, the system offers an environment for trainees to practice core competencies essential to bakery sales. An initial user study with ten participants offered valuable insights into the system's usability, user experience, and perceived effectiveness in supporting skill acquisition and retention within a virtual training environment. Usability assessments, reflected in an average System Usability Scale (SUS) score of 72.5, further indicated a favorable reception of the platform.

While the current VR training system demonstrates significant potential, several areas for future development could further enhance its effectiveness. A primary objective is the integration of adaptive feedback mechanisms that deliver real-time corrective guidance during task execution, enabling trainees to address errors immediately and thereby enhancing the effectiveness of the learning process. Additionally, introducing challenging, high-stress scenarios would enable trainees to practice prioritizing tasks under pressure, thereby strengthening decision-making skills critical for real-world applications. In order to make the system more realistic and relevant to bakery settings, additional steps could be incorporated.

REFERENCES

- [1] J. Brooke et al. Sus-a quick and dirty usability scale. *Usability evaluation in industry*, 189(194):4–7, 1996. 2
- [2] F.-K. Chiang, X. Shang, and L. Qiao. Augmented reality in vocational training: A systematic review of research and applications. *Computers in Human Behavior*, 129:107125, 2022. 1
- [3] F. Farzinnejad, J. Rasti, N. Khezrian, and J. Grubert. The effect of an exergame on the shadow play skill based on muscle memory for young female participants: The case of forehand drive in table tennis. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 1152–1160. IEEE, 2023. 1
- [4] J. Spilski, J.-P. Exner, M. Schmidt, A. Makhkamova, S. Schlittmeier, C. Giehl, T. Lachmann, M. Pietschmann, and D. Werth. Potential of vr in the vocational education and training of craftsmen. In *Proceedings of the 19th International Conference on Construction Applications of Virtual Reality*, pp. 181–190, 2019. 1
- [5] H. Stefan, M. Mortimer, and B. Horan. Evaluating the effectiveness of virtual reality for safety-relevant training: a systematic review. *Virtual Reality*, 27(4):2839–2869, 2023. 1
- [6] H. M. Tusher, S. Mallam, and S. Nazir. A systematic review of virtual reality features for skill training. *Technology, Knowledge and Learning*, 29(2):843–878, 2024. 1